A Reproducible Deep Reinforcement Learning Framework for Single-Intersection Traffic Signal Control under Multi-Scenario Evaluation

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Abstract

We propose a reinforcement learning framework for adaptive traffic signal control at a single intersection using the SUMO simulator. Our environment features state representations with queue dynamics, temporal representations, and pedestrian models for morning-peak and off-peak traffic scenarios. We mainly focus on Double Dueling DQN, and compare its performance with fixed-time control, longest-queue-first, Q-learning, and SARSA. Experimental results show that DRL methods consistently reduce queue lengths and delays while improving traffic throughput. The framework provides a scalable and reproducible experimental environment for future research on multi-intersection coordination and mixed-traffic signal control.

1. Introduction

Traffic congestion remains a major challenge for urban mobility, highlighting the importance of effective traffic signal control at intersections. Traditional signal control methods are often designed under simplified assumptions, struggling to cope with dynamic and uncertain traffic conditions. With the growing availability of traffic data and computational resources, recent advances in deep reinforcement learning (DRL) enable agents to learn control policies directly through interaction with complex traffic environments, making them effective for capturing nonlinear and high-dimensional traffic dynamics. In this context, we study single-intersection traffic signal control in a SUMO simulation that incorporates realistic traffic dynamics, temporal features and pedestrian interactions, providing a reproducible framework for evaluating DRL-based approaches.

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Code Repository: https://github.com/Edward67795/traffic_rl_project

2. Related Work and Novel Contribution

2.1. Traditional Signal Control and Heuristic Methods

Traditional traffic signal control mainly based on fixed-time plans that are simple and stable but lack responsiveness to real-time traffic demand. (Hunt et al., 1981). Later developed adaptive methods adjust cycle lengths and green splits using detector feedback, improving responsiveness but still relying heavily on hand-crafted rules (El-Tantawy et al., 2013). Rule-based strategies such as Longest-Queue-First and Max-Pressure are more reactive to instantaneous traffic states (Wunderlich, 2007; Varaiya, 2013), yet they struggle to optimize long-term performance and often have difficulties in complex or highly dynamic conditions.

2.2. Reinforcement Learning for Traffic Signal Control

Reinforcement learning (RL) has emerged as a promising direction for traffic signal control due to its self-learning and adaptive capabilities. Early tabular RL methods such as Q-learning and SARSA rely on discrete state representations, making them difficult to scale to real-world traffic scenarios (Dayan & Watkins, 1992; Rummery & Niranjan, 1994). Deep reinforcement learning (DRL) leverages neural networks to process continuous and high-dimensional information. Variants such as DQN have demonstrated strong results in both single-intersection and multi-agent settings (Mnih et al., 2015; Van Hasselt et al., 2016; Wang et al., 2016; Schaul et al., 2015). More recent research explores richer state designs, including temporal stacking, arrival-rate estimation, and more expressive reward functions (Wei et al., 2019). However, differences in environment design and reward shaping make fair comparisons difficult, which is why standardized frameworks are needed.

2.3. SUMO-Based Simulation and Reproducible Frameworks

SUMO (Simulation of Urban Mobility) is widely used in traffic signal control research due to its flexibility, detailed vehicle-level data, and suitability (Krajzewicz et al., 2012). A clear and standardized framework is essential for DRL-based traffic control (Wu et al., 2017). However, differences between studies make reproducibility difficult.

2.4. Novel Contributions.

We present a reproducible and extensible DRL-based traffic signal control framework with four key contributions: (i) an enhanced RL environment state representations with queue dynamics, temporal features and a pedestrian model; (i) realistic morning-peak and off-peak scenarios with varied traffic demands; (iii) comprehensive baselines covering DRL and classical methods; and (iv) a unified and modular pipeline for training, evaluation, and visualization.

3. Approach: algorithm development

3.1. Environment

3.1.1. Problem Formulation

We formulate isolated intersection traffic signal control as a Markov Decision Process (MDP) $\langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$, where the agent observes state s_t, selects action a_t, and receives reward r_t at each time step. We consider a single four-leg intersection with traffic dynamics governed by SUMO, and assume approximate Markovianity at the control time scale through appropriate state design.

3.1.2. Action Space

The action space is discrete, $\mathcal{A} = \{0, 1\}$, corresponding to selecting the green phase for the north–south or east–west direction. Phase transitions, including yellow phases and minimum green-time constraints, are handled by the environment to ensure safety and avoid unrealistic switching.

3.1.3. State Representation

The state is a fixed-dimensional continuous vector formed by concatenating vehicular traffic states, pedestrian states, temporal dynamics, traffic demand estimates, and signal phase information.

Lane-Level Traffic State For each incoming lane, the state includes normalized measures of vehicle queue length, accumulated waiting time, and lane occupancy, capturing the instantaneous traffic load at the intersection.

Temporal Dynamics To capture short-term traffic evolution, we include first-order temporal differences of queue length and waiting time for each incoming lane:

$$\Delta q_t = \frac{q_t - q_{t-1}}{\Delta t}, \qquad \Delta w_t = \frac{w_t - w_{t-1}}{\Delta t}, \tag{1}$$

where Δt denotes the control interval. These features provide explicit trend information without requiring recurrent models.

Arrival Rate Estimation Short-term vehicle arrival rates are estimated from recent changes in lane-level vehicle counts, providing the agent with predictive information about near-future traffic demand and mitigating partial observability.

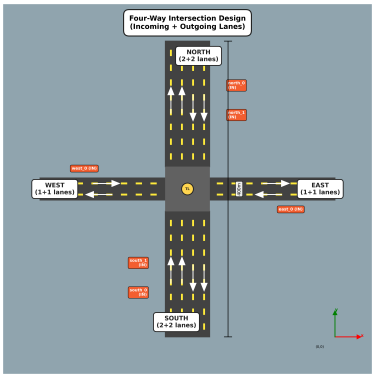


Figure 1. Four-way intersection.

Signal Phase Encoding The current signal phase is represented using a one-hot encoding, together with the normalized elapsed duration of the active phase. This allows the agent to jointly reason about right-of-way and phase timing.

Pedestrian State Modeling Pedestrian traffic is explicitly modeled for both the north–south and east–west directions. The corresponding pedestrian queue lengths and accumulated waiting times are incorporated into the state, enabling joint optimization of vehicular efficiency and pedestrian delay.

3.1.4. Reward Function

The reward is defined as a weighted sum of operational costs:

$$\begin{aligned} r_t = & -w_q \cdot \frac{Q_t}{Q_{\max}} - w_w \cdot \frac{W_t}{W_{\max}} - w_e \cdot \frac{E_t}{E_{\max}} \\ & - w_p \cdot \frac{P_t}{P_{\max}} - w_s \cdot \mathbb{I}(a_t \neq a_{t-1}). \end{aligned} \tag{2}$$

where Q_t, W_t, E_t, and P_t denote total vehicle queue length, vehicle waiting time, CO_2 emissions(provided by SUMO), and pedestrian waiting time, respectively. The indicator function penalizes phase switching to encourage smooth and stable control. All terms are normalized to ensure comparable scales.

3.1.5. Environment Dynamics

Each selected action is applied for a fixed control interval during which the traffic simulator advances according to microscopic dynamics. An episode terminates after a predefined simulation horizon corresponding to a complete traffic demand cycle.

3.2. Comparative Baseline Methods

To evaluate our reinforcement learning approach, we implement representative rule-based and tabular reinforcement learning baselines within the same SUMO environment.

Fixed-Time Control Fixed-Time Control uses predetermined green durations for two alternating phases serving the north–south and east–west directions. Lacking adaptivity to real-time traffic conditions, it often yields suboptimal performance.

Longest-Queue-First Control The Longest-Queue-First (LQF) controller selects the phase with the larger queue, choosing Phase 0 when $queue_{ns} \geq queue_{ew}$ and Phase 1 otherwise. While adaptive, LQF is myopic and lacks long-term optimization.

Q-learning The tabular Q-learning controller selects actions by maximizing state–action values over a discretized state space with limited states, defined by the current phase and coarse queue conditions. While capable of learning from reward feedback, the coarse discretization limits its expressiveness and long-term performance.

SARSA Expected SARSA is used as a tabular reinforcement learning baseline that updates Q-values using policy-weighted expectations, leading to smoother learning than Q-learning but limited by discretized state representations.

3.3. Deep Q-Network Methods

Traditional reinforcement learning methods rely on discrete state representations, limiting their ability to model continuous traffic dynamics. We therefore adopt a Deep Q-Network (DQN) to approximate the action-value function and further incorporate Double and Dueling architectures, resulting in a Double Dueling DQN for improved value estimation and training stability.

Deep Q-Network (DQN) The vanilla DQN approximates the action-value function $Q(s, a)$ using a deep neural network that maps continuous states to Q-values for all actions. To improve training stability, DQN employs experience replay, which samples transitions (s_t, a_t, r_t, s_{t+1}) from a replay buffer to reduce temporal correlations, and a target network that is periodically updated from the online network. Despite these stabilization mechanisms, naive DQN still suffers from overestimation bias because the same network is used for both action selection and value evaluation.

Double DQN Double DQN reduces overestimation by decoupling action selection from action evaluation. The online network selects the action, while the target network evaluates its value. The target is computed as

$$y_t^{\mathrm{DDQN}} = r_t + \gamma Q\Big(s_{t+1}, \arg\max_{a'} Q(s_{t+1}, a' \mid \theta) \;\Big|\; \theta^-\Big). \tag{3}$$

The loss function minimized during training is

$$L(\theta) = \mathbb{E}\left[\Big(y_t^{\mathrm{DDQN}} - Q(s_t, a_t \mid \theta)\Big)^2\right]. \tag{4}$$

This modification improves value estimation reliability without increasing network complexity.

Dueling DQN Dueling DQN enhances the representational capacity of the Q-network by decomposing the action-value function into a state-value function and an advantage function. Given a shared feature representation, the network estimates

$$V(s) = V(s \mid \theta_V), \qquad A(s, a) = A(s, a \mid \theta_A). \tag{5}$$

To ensure identifiability, the Q-value is aggregated as

$$Q(s, a) = V(s) + \left(A(s, a) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s, a')\right). \tag{6}$$

This structure allows the network to efficiently learn state values when differences among action returns are small, which frequently occurs in traffic signal control tasks.

Double Dueling DQN The Double Dueling DQN (DDDQN) architecture combines the target computation of Double DQN with the dueling network structure. The network consists of shared feature extraction layers followed by separate value and advantage streams.

The target value is defined as

$$y_t^{\mathrm{DDDQN}} = r_t + \gamma Q\Big(s_{t+1}, \arg\max_{a'} Q(s_{t+1}, a' \mid \theta) \;\Big|\; \theta^-\Big). \tag{7}$$

The Q-value produced by the target network is

$$\begin{aligned} Q(s, a \mid \theta^-) = V(s \mid \theta_V^-) + \Big(&A(s, a \mid \theta_A^-) \\ &- \frac{1}{|\mathcal{A}|}\sum_{a'} A(s, a' \mid \theta_A^-)\Big). \end{aligned} \tag{8}$$

The online network follows the same dueling structure, and training minimizes

$$L(\theta) = \mathbb{E}\left[\Big(y_t^{\mathrm{DDDQN}} - Q(s_t, a_t \mid \theta)\Big)^2\right]. \tag{9}$$

By reducing overestimation bias and improving state representation efficiency, the proposed architecture is well suited for traffic signal control tasks requiring stable and accurate value estimation.

4. Experiment Results

4.1. Experimental Setup

All experiments use the SUMO traffic simulator with the four-way intersection configuration described in Section 3. Two traffic scenarios are evaluated: Morning Rush with heavy directional flow and Off-Peak with balanced light traffic. Each method is evaluated over 5 independent runs per scenario. Performance metrics include average queue length, total episode reward, maximum queue (peak congestion), and throughput. Two experimental conditions are tested: vehicle-only control (39-dimensional state) and mixed traffic with virtual pedestrians (43-dimensional state).

4.2. Training Configuration

Training proceeds for 400 episodes, each spanning 3,600 simulated seconds with 5-second decision intervals. The replay buffer stores 100,000 transitions, from which mini-batches of 128 are sampled for gradient updates using Adam optimizer with learning rate 3×10^{-4}. The target network synchronizes every 500 steps. Exploration follows an ε-greedy policy with ε decaying linearly from 1.0 to 0.01 over 300 episodes. The discount factor is set to $\gamma = 0.99$.

4.3. Vehicle-Only Performance

Table 1 presents results for the vehicle-only scenario. In the Morning Rush scenario, Dueling Double DQN achieves an average queue of 2.99 vehicles compared to 10.45 for Fixed-60s, representing a 71% reduction. The Longest-Queue-First (LQF) heuristic achieves comparable performance (2.86 average queue), slightly outperforming DQN. This result reflects that queue minimization aligns directly with LQF's greedy objective, and single-intersection control lacks the coordination complexity where learned policies typically excel.

DQN outperforms both tabular methods by approximately 20% in queue length. Tabular Q-learning and SARSA achieve average queues of 3.79 and 3.68 respectively, compared to 2.99 for DQN. This improvement stems from the neural network's ability to generalize across continuous 39-dimensional states, while tabular methods rely on coarse 18-state discretization that loses information about fine-grained traffic dynamics.

4.4. Pedestrian Scenario Performance

When virtual pedestrians are introduced, the control problem becomes significantly more challenging. The reward function now incorporates pedestrian waiting time penalties, and the agent must balance vehicle throughput against pedestrian service. Table 2 presents results for this mixed-traffic scenario.

Table 1. Vehicle-only performance comparison. Best values per scenario in **bold**. DQN refers to Dueling Double DQN.

	Morning Rush		Off-Peak	
Method	Avg Q	Reward	Avg Q	Reward
DQN (Ours)	2.99	**-233.5**	1.58	-134.5
LQF	**2.86**	-228.2	**1.26**	**-124.0**
SARSA	3.68	-256.2	1.64	-134.6
Q-Learning	3.79	-260.4	1.69	-135.6
Fixed-30s	5.19	-311.0	1.85	-141.6
Fixed-60s	10.45	-481.8	3.50	-188.6

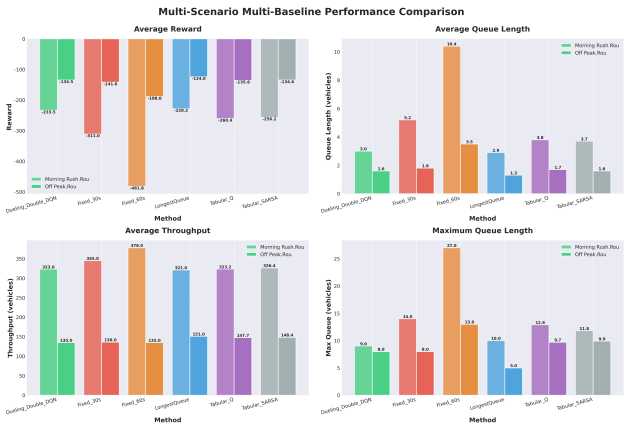


Figure 2. Vehicle-only scenario performance comparison. DQN and LQF achieve comparable results, both significantly outperforming fixed-time control.

The advantage of deep reinforcement learning becomes substantially more pronounced in this setting. DQN achieves an average queue of 2.80 in Morning Rush, while tabular methods degrade to 6.20 (Q-Learning) and 6.50 (SARSA)—an increase of over 120%. This degradation occurs because the 18-state tabular representation cannot capture the additional pedestrian state dimensions, forcing these methods to ignore critical information. In contrast, DQN seamlessly incorporates the 4 additional pedestrian features into its 43-dimensional state representation.

Notably, even LQF's relative advantage diminishes in the pedestrian scenario. While LQF matched DQN performance in vehicle-only control, it achieves only -437.7 reward compared to DQN's -418.4 in Morning Rush with pedestrians. LQF's greedy queue-based decisions cannot account for pedestrian waiting times, whereas DQN learns to balance both objectives through the composite reward signal.

4.5. Training Dynamics

Figure 4 presents training curves over 400 episodes. In the vehicle-only scenario (top row), reward improves from approximately -300 to -240 (20% gain) while average queue

Table 2. Pedestrian scenario performance. DQN demonstrates clear advantages when pedestrian coordination is required.

	Morning Rush		Off-Peak	
Method	Avg Q	Reward	Avg Q	Reward
DQN (Ours)	**2.80**	**-418.4**	**1.00**	**-341.3**
LQF	2.90	-437.7	1.30	-420.3
SARSA	6.50	-688.3	2.00	-474.5
Q-Learning	6.20	-616.5	2.30	-553.4
Fixed-30s	5.20	-701.1	1.80	-561.1
Fixed-60s	10.40	-1175.9	3.50	-851.1

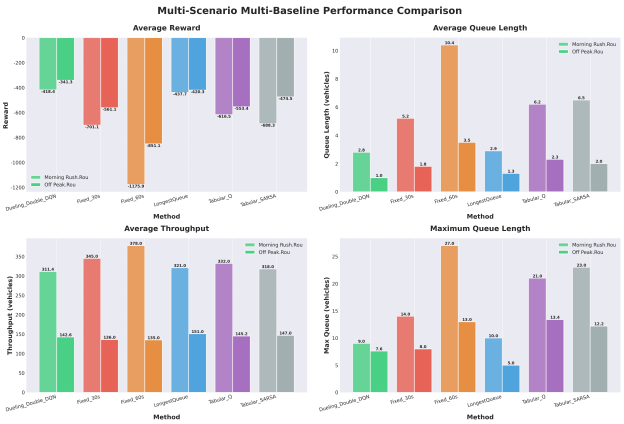


Figure 3. Pedestrian scenario performance comparison. DQN maintains strong performance while tabular methods degrade significantly (120%+ increase in queue length).

length decreases from over 5 to 3.2 vehicles, with convergence around episode 150. The pedestrian scenario (bottom row) exhibits a more challenging trajectory: rewards start at approximately -600 due to pedestrian waiting time penalties, improving to -460 after convergence. Queue length decreases from over 6 to approximately 3.5 vehicles, with longer convergence time (episode 200) reflecting the increased complexity of multi-objective optimization.

The temporal derivative features contribute to efficient learning by providing gradient signals that help attribute outcomes to preceding state changes. Analysis of the learned policy reveals preference for extending green phases during high-arrival-rate periods. The queue derivative enables anticipatory control: the policy tends to switch phases when $\Delta q > 0$ (building congestion) even when current queues remain moderate.

4.6. Analysis and Discussion

The experimental results highlight the value of temporally enhanced state representations and the conditions under which deep RL excels. The 39/43-dimensional state space incorporating queue derivatives and arrival rates enables

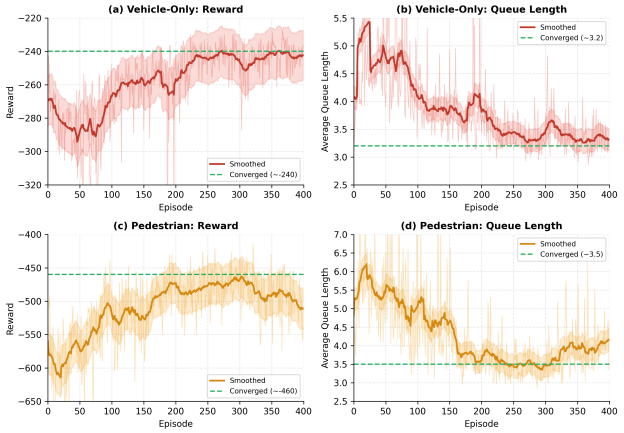


Figure 4. Training curves for vehicle-only (top) and pedestrian (bottom) scenarios over 400 episodes. (a,c) Reward convergence showing 20% improvement. (b,d) Queue length reduction from initial exploration to stable policy.

anticipatory control strategies, allowing the policy to detect congestion trends before they become severe and extend green phases during high-arrival-rate periods. In simple vehicle-only scenarios, the interpretable LQF heuristic performs comparably to DQN, achieving even slightly lower queue lengths (2.86 vs 2.99), as its greedy objective aligns well with single-intersection queue minimization. However, when pedestrian coordination introduces multi-objective optimization requirements, DQN demonstrates clear advantages: it achieves -418.4 reward compared to LQF's -437.7 by learning to balance vehicle throughput and pedestrian service through the composite reward signal. This suggests that problem complexity is the key determinant for algorithm selection, which means learned policies provide the greatest benefit when control objectives become multi-dimensional or when coordination across competing demands is required.

5. Conclusion

This paper presented a deep reinforcement learning framework for single-intersection traffic signal control using temporally enhanced state representations. The proposed Dueling Double DQN with 39/43-dimensional state space achieves 71% queue reduction compared to fixed-cycle control and outperforms tabular methods by 20%. The introduction of pedestrians reveals the scalability advantage of deep RL that DQN seamlessly incorporates additional state dimensions while tabular methods degrade by over 120%. Future work will extend this framework to multi-intersection coordination and integration with real-world traffic data.

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Statement of Contributions

Zehao Li (zl3667): Designed and implemented the SUMO-based simulation environment, including intersection network construction, traffic scenario modeling (morning rush and off-peak routes). Initiated the training pipeline and integrated other features developed by teammates. Performed quantitative result analysis, generated performance comparison visualizations and training curve figures, and authored the experimental results and conclusion sections of the paper (Sections 4 and 5).

Yizheng Lin (yl6097): Designed and implemented an early-stage environment and an initial DQN-based model, both developed independently of the SUMO simulator. Integrated baseline methods and pedestrian modeling into the SUMO-based simulation environment, and contributed to the refinement of the proposed approach. Authored the Abstract, Introduction, Related Work and Novel Contribution, as well as the baseline and DQN components of the Approach section, and refined the environment description based on team contributions.

Weihao Zhou (wz2750): Enhanced the SUMO environment with temporal features including derivative calculations for queue and waiting time change rates, arrival rate estimation, wrote the Environment section. Conducted hyperparameter tuning and environment parameter optimization. Performed comprehensive literature review on reinforcement learning approaches for traffic signal control, collecting citations and establishing theoretical foundations, and refined the Related Work section.

Code Repository

https://github.com/Edward67795/traffic_rl_project